are generally composed of at least 10 million billion billion individual particles. If each one of those particles' wave functions is compulsively pulling the handle of its own slot machine (Figure 10.3b), then, on average, at least one of them will hit the collapse jackpot every millionth of a second. But because the particles in this book are all continually interacting with each other, they're all entangled—which means they all share a single wave function. So when one of them hits the jackpot, the wave function for the entire book collapses, meaning this book can't be in two places at once for much longer than a microsecond or so—a hundred thousand times faster than a blink of an eye. As Bell put it, in spontaneous-collapse theory, Schrödinger's cat "is not both dead and alive for more than a split second." This neatly resolves the measurement problem: all objects, large and small, obey the same laws, with no special role played by measurement. Wave function collapse happens at random to everything, all the time, without any need for intervention from an observer.

Spontaneous-collapse theory isn't really just one theory; it's a collection of related theories, which were developed by a small handful of people dissatisfied with the Copenhagen interpretation over the years. The one described above—the one that caught Bell's attention and through him the attention of many other physicists—was developed in 1985 by a trio of physicists working in Italy, GianCarlo Ghirardi, Alberto Rimini, and Tullio Weber. It came to be known as the "GRW model" after their initials. "I see the GRW model as a very nice illustration of how quantum mechanics, to become rational, requires only a change which is very small (on some measures!)," Bell wrote soon after the GRW paper first appeared. Bell's paper on GRW brought it to the attention of many other physicists, including Philip Pearle, who had been working on similar ideas since the early 1970s. (Ten years earlier, Pearle's work had also singled him out for the unfortunate distinction of an interview with a sociologist studying "social deviance" among physicists.) Pearle wrote to Bell for more information on GRW, and Bell arranged for Pearle to spend a sabbatical with Ghirardi, where the two men attempted to adapt GRW for relativistic quantum field theory. But the common refrain that had greeted Bohm, Everett, and others for decades hit GRW and Pearle doubly hard: quantum theory works astonishingly well. Why fix what is manifestly not broken? Why do we even need a different interpretation, much less a different theory